



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Extension 1

Assessment Task 3

Term 2, 2012

Name: _____ Mathematics Class: _____

Time Allowed: 50 minutes + 2 minutes reading time

Total Marks: 38

Instructions:

- Attempt all eight questions.
- Show all necessary working in questions 6 to 8.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.
- A table of standard integrals is provided at the back of this paper.

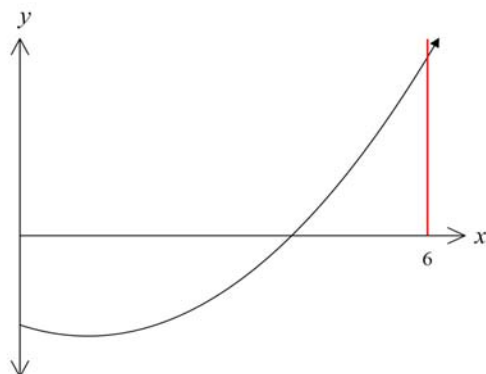
Question	1-1,2	1-3,4,5	2-6abc	2-6d	2-6e	2-7a	2-7b	2-7c	2-8a	2-8b	2-8c	Total
PE3		/3				/4			/2			/9
HE2					/3					/3		/6
HE6				/3				/4				/7
HE7	/2		/5				/4				/5	/16
												/38

SECTION 1. (5 marks - 1 mark each)

Multiple choice.

Write down the correct alternative on the answer sheet provided.

1. The diagram below shows the graph of $y = x^2 - 2x - 8$.



What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$?

- (A) $A = \int_0^5 (x^2 - 2x - 8)dx + \left| \int_5^6 (x^2 - 2x - 8)dx \right|$
- (B) $A = \int_0^4 (x^2 - 2x - 8)dx + \left| \int_4^6 (x^2 - 2x - 8)dx \right|$
- (C) $A = \left| \int_0^5 (x^2 - 2x - 8)dx \right| + \int_5^6 (x^2 - 2x - 8)dx$
- (D) $A = \left| \int_0^4 (x^2 - 2x - 8)dx \right| + \int_4^6 (x^2 - 2x - 8)dx$
2. $\lim_{x \rightarrow 0} \frac{\sin 3x \cos y}{4x} =$

- (A) $\frac{4}{3}$ (C) $\frac{3}{4}$
- (B) $\frac{4 \cos y}{3}$ (D) $\frac{3 \cos y}{4}$

3. How many numbers greater than 3 000 can be formed from the digits 2, 3, 4, 5 and 6 if no digit is repeated?

- (A) 96 (C) 196
- (B) 120 (D) 216

4. How many distinct permutations of the letters of the word 'ATTAINS' are possible when the word begins and ends with the letter T?
- (A) 60 (C) 120
(B) 360 (D) 1260
5. Eden, Toby and four friends arrange themselves at random in a circle. What is the probability that Eden and Toby are *not* together?
- (A) $\frac{1}{120}$ (C) $\frac{2}{5}$
(B) $\frac{3}{5}$ (D) $\frac{119}{120}$

SECTION 2. (33 marks)

Marks

- Answer each question in a separate booklet.
- Show all necessary working.

Question 6. (11 marks)

- (a) Find $\int \frac{e^x}{1+e^x} dx$ **1**
- (b) Use the table of standard integrals to evaluate $\int_0^{\frac{\pi}{8}} \sec 2x \tan 2x dx$ **2**
- (c) Find $\frac{d}{dx}(x \sin^2 x)$ **2**
- (d) Find $\int \frac{x^2 dx}{(x-2)^2}$ using the substitution $u = x-2$ **3**
- (e) Prove by mathematical induction that $3^{2n} - 1$ is divisible by 8 for all positive integers n . **3**

Question 7. (12 marks)

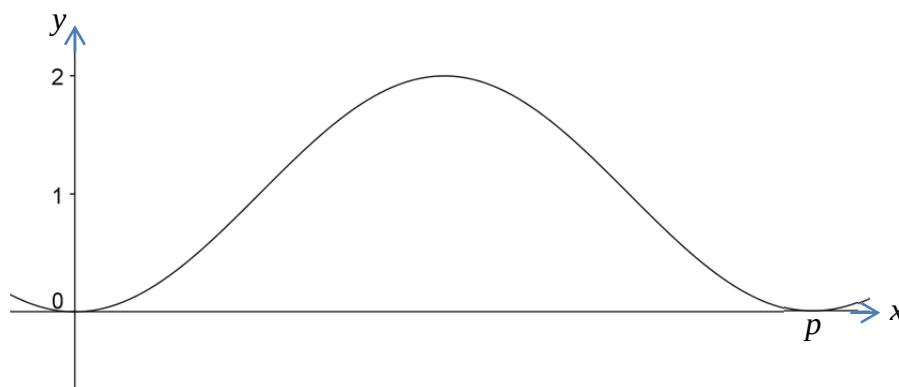
- (a) A team of five contestants for a quiz game is to be selected at random from a class of eight girls and five boys.

- (i) What is the probability the team consists of 3 girls and 2 boys? **2**
(ii) Find the probability there are more girls chosen than boys? **2**
(Give both answers correct to 2 decimal places.)

- (b) Consider the function $y = x^2 e^{2x}$

- (i) Show that $\frac{dy}{dx} = 2xe^{2x}(1+x)$ **1**
(ii) Find any stationary points and their nature. **3**
(iii) Sketch $y = x^2 e^{2x}$. **1**

- (c)

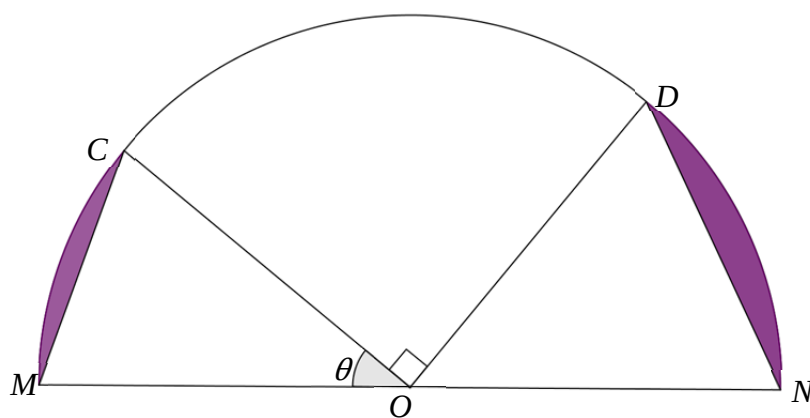


The diagram above shows part of the graph of $y = 1 - \cos 2x$.

- (i) Write down the value of p . **1**
(ii) The region between $y = 1 - \cos 2x$, the x -axis, $x = 0$ and $x = p$ is rotated about the x -axis. Find the volume of the solid generated. **3**

Question 8. (10 marks)

- (a) Three girls and w boys are arranged randomly in a line. Find the probability that the girls are together. Give your answer as a simplified algebraic fraction in terms of w . 2
- (b) Use the principle of mathematical induction to show that $4^n - 1 - 7n > 0$ for all integers $n \geq 2$. 3
- (c)



In the diagram MN is the diameter of a semicircle centre O and radius 1 metre.

C and D are variable points that move on the semicircle such that

$$\angle MOC = \theta \text{ and } \angle COD = 90^\circ.$$

- (i) Show that the total area of the shaded segments A is given by

$$A = \frac{\pi}{4} - \frac{1}{2}(\sin \theta + \cos \theta) \quad \mathbf{2}$$

- (ii) Hence find the minimum shaded area A in exact form. **3**

[illegible]

End of test

NSGHS Task3 X1 Maths 2012 Solutions

SECTION 1

1. x-intercept: $x^2 - 2x - 8 = 0$
 $(x - 4)(x + 2) = 0$
 $x = 4, -2$

Area between $x = 0$ and $x = 4$ negative

$$A = \left| \int_0^4 (x^2 - 2x - 8) dx \right| + \int_4^6 (x^2 - 2x - 8) dx$$

Correct answer **(D)**

2. $\lim_{x \rightarrow 0} \frac{\sin 3x \cos y}{4x} =$

$$\begin{aligned} & \cos y \times \frac{3}{4} \times \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \\ &= \cos y \times \frac{3}{4} \times 1 \\ &= \frac{3 \cos y}{4} \end{aligned}$$

Correct answer **(D)**

3. 5 digit numbers = $5! = 120$
4 digit numbers = $4 \times 4! = 96$
Answer = $120 + 96 = 216$
Correct answer **(D)**

4. T(aains)T
 $1 \times (5! \div 2!) \times 1$ ways
 $= 120 \div 2 = 60$ ways
Correct answer **(A)**

5. E, T, A, B, C, D can form a ring of 6 in $5!$ ways - $n(S)$
Restriction- seat E = 1 way
seat T = 3 ways
seat rest = $4!$ Ways
 $P(\text{not together}) = (1 \times 3 \times 4!) \div 5!$
 $= 3 \div 5$
Correct answer **(B)**

or Regardless of where E sits, T has a choice of 5 seats, 3 of which are not next to E. i.e. $P(\text{not together}) = 3 \div 5$

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SECTION 2

Question 6.

(a) $\int \frac{e^x}{1 + e^x} dx$
 $= \ln(1 + e^x) + c$

(b) $\int_0^{\frac{\pi}{8}} \sec 2x \tan 2x dx$
 $= \frac{1}{2} [\sec 2x]_0^{\frac{\pi}{8}}$
 $= \frac{1}{2} [\sec(\frac{\pi}{4}) - \sec(0)]$
 $= \frac{1}{2} (\sqrt{2} - 1)$

(c) $\frac{d}{dx}(x \sin^2 x)$
 $= 1 \times \sin^2 x + x \times [2 \sin x \times \frac{d}{dx}(\sin x)]$
 $= \sin^2 x + 2x \sin x \cos x$
 $= \sin x(\sin x + 2x \cos x)$

(d) $\int \frac{x^2 dx}{(x-2)^2} =$
let $u = x - 2$
 $\therefore \frac{du}{dx} = 1$
 $du = dx, x = u + 2$
 $\int \frac{x^2 dx}{(x-2)^2} = \int \frac{(u+2)^2 du}{u^2}$
 $= \int \frac{(u^2 + 4u + 4) du}{u^2}$
 $= \int (1 + \frac{4}{u} + 4u^{-2}) du$
 $= u + 4 \ln u - \frac{4}{u} + c$
 $+ x - 2 + 4 \ln(x-2) - \frac{4}{x-2} + c$

(e)

When $n = 1$, $3^{2n} - 1 = 9 - 1 = 8$

Statement true for $n = 1$

Now assume statement true for $n = k$,

$0 \leq k \leq n$

i.e. $3^{2k} - 1 = 8J$, J integer

$$\therefore 3^{2k} = 8J + 1$$

need to show statement true for next term

i.e. $n = k + 1$

When $n = k + 1$, $3^{2n} - 1$

$$= 3^{2(k+1)} - 1$$

$$= 3^{2k+2} - 1$$

$$= 3^2 \times 3^{2k} - 1$$

$$= 9 \times (8J + 1) - 1$$

$$= 72J + 8$$

$$= 8(9J + 1)$$

Which is divisible by 8 (common factor)

true for $n = k + 1$ when true for $n = k$

Since true for $n = 1$ and true for next term,
true for all positive integers n .

Question 7.

(a) Select 5 from 8 girls, 5 boys

${}^{13}C_5$ ways

(i) Choose 3 girls = 8C_3 ways

Choose 2 boys = 5C_2 ways

$$\begin{aligned} P(3 \text{ girls, } 2 \text{ boys}) &= \frac{{}^8C_3 \times {}^5C_2}{{}^{13}C_5} \\ &= 0.44 \text{ (2 dp)} \end{aligned}$$

(ii) $P(\text{More girls than boys}) =$
 $(3G, 2B) \text{ or } (4G, 1B) \text{ or } (5G)$

$$= \frac{{}^8C_3 \times {}^5C_2 + {}^8C_4 \times {}^5C_1 + {}^8C_5}{{}^{13}C_5}$$

$$= 0.75 \text{ (2 dp)}$$

(b)

(i)

$$y = x^2 e^{2x}$$

$$\begin{aligned} \frac{dy}{dx} &= 2x \times e^{2x} + 2e^{2x} \times x^2 \\ &= 2xe^{2x}(1 + x) \end{aligned}$$

(iii) For stationary points $\frac{dy}{dx} = 0$

$$2xe^{2x}(1 + x) = 0$$

$$\therefore 2x = 0, 1 + x = 0, e^{2x} \neq 0$$

$$x = 0, y = 0$$

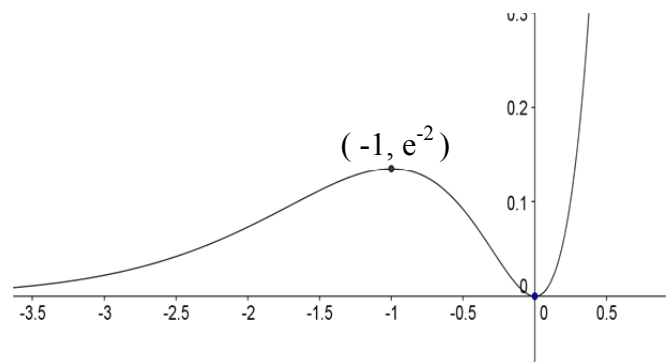
$$x = -1, y = (-1)^2 e^{-2} = \frac{1}{e^2}$$

x	-2	-1	-0.5	0	1
y	$4e^{-2}$	0	$-0.5e$	0	$4e^2$
	+	0	-	0	+

Maximum at $(-1, e^{-2})$

Minimum at $(0, 0)$

(iii)



(c)

(i)

$$1 - \cos 2x = 0$$

$$\cos 2x = 1$$

$$2x = 0, 2\pi$$

$$x = \pi$$

$$p = \pi \text{ (first positive value)}$$

(ii)

$$\begin{aligned} V &= \pi \int_0^{\pi} (1 - \cos 2x)^2 dx \\ &= \pi \int_0^{\pi} (1 - 2\cos 2x + \cos^2 2x) dx \\ &= \pi \int_0^{\pi} [1 - 2\cos 2x + \frac{1}{2}(1 + \cos 4x)] dx \\ &= \pi \left[x - \sin 2x + \frac{x}{2} + \frac{1}{8} \sin 4x \right]_0^{\pi} \\ &= \pi \left[\left(\pi - \sin 2\pi + \frac{\pi}{2} + \frac{1}{8} \sin 4\pi \right) - (0 - \sin 0 + 0 - \sin 0) \right] \\ &= \pi \left[\left(\pi + \frac{\pi}{2} \right) - 0 \right] \\ &= \frac{3\pi^2}{2} \end{aligned}$$

Question 8.

(a) 3 girls, w boys in line
no restriction : $(w + 3)!$ ways
girls together: $3!$ ways,
line of $(w + 1) = (w + 1)!$ ways
P(girls together)
$$= \frac{(w+1)! \times 3!}{(w+3)!}$$
$$= \frac{6}{(w+3)(w+2)}$$

(b)

Prove $4^n - 1 - 7n > 0$ for $n \geq 2$

When $n = 2$, LHS = $4^2 - 1 - 7 \times 2 = 1$

True for $n = 2$

Assume true for $n = k$

i.e. $4^k - 1 - 7k > 0$

to show true for next term

i.e. $4^{k+1} - 1 - 7(k+1) > 0$

LHS = $4^{k+1} - 1 - 7(k+1)$

$$= 4 \times 4^k - 1 - 7k - 7$$

$$= 4 \times 4^k - 7k - 8$$

$$= 4(4^k - 1 - 7k) + 21k - 4 > 0$$

Since $4^k - 1 - 7k > 0$ by assumption

$21k - 4 > 0$ since $k \geq 2$

True for next term $n = k + 1$

true for $n = k + 1$ when true for $n = k$

Since true for $n = 2$ and true for next term,
true for all positive integers $n \geq 2$.

(c)

(i) $\angle NOD = \frac{\pi}{2} - \theta$

$$\begin{aligned} \therefore A &= \frac{1}{2} \times 1^2 (\theta - \sin \theta) + \frac{1}{2} \times 1^2 \left[\left(\frac{\pi}{2} - \theta \right) - \sin \left(\frac{\pi}{2} - \theta \right) \right] \\ &= \frac{1}{2} [\theta - \sin \theta + \frac{\pi}{2} - \theta - \cos \theta] \\ &= \frac{1}{2} [-\sin \theta + \frac{\pi}{2} - \cos \theta] \\ &= \frac{\pi}{4} - \frac{1}{2} (\sin \theta + \cos \theta) \end{aligned}$$

(ii)

$$\frac{dA}{d\theta} = 0 - \frac{1}{2} (\cos \theta - \sin \theta)$$

$$= \frac{1}{2} (\sin \theta - \cos \theta)$$

$$\frac{d^2 A}{d\theta^2} = \frac{1}{2} (\cos \theta + \sin \theta)$$

Stationary points when

$$\frac{dA}{d\theta} = \sin \theta - \cos \theta = 0$$

$$\tan \theta = 1$$

$$\theta = \frac{\pi}{4} \quad (\theta \text{ acute})$$

$$\frac{d^2 A}{d\theta^2} = \cos \frac{\pi}{4} + \sin \frac{\pi}{4} = \frac{2}{\sqrt{2}} > 0$$

Concave up. Minimum at $x = \frac{\pi}{4}$

$$A_{\min} = \frac{\pi}{4} - \frac{1}{2} \left[\sin \left(\frac{\pi}{4} \right) + \cos \left(\frac{\pi}{4} \right) \right]$$

$$= \frac{\pi}{4} - \frac{1}{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right)$$

$$= \frac{\pi}{4} - \frac{1}{\sqrt{2}}$$

